CHEM-1310
Fall 2017
Second Examination
Form A

Multiple Choice - Circle the BEST Answer

1. Which of the following molecules in the gas phase will have the slowest rate of effusion?

   A) F_2
   B) Br_2
   C) CO_2
   D) CH_4
   E) All molecules have the same rate of effusion.

2. Which of the following statement(s) about the kinetic-molecular theory is/are true?

   A) The kinetic-molecular theory describes that when molecules collide their collisions are elastic.
   B) The kinetic-molecular theory describes how gas molecules are so tiny that they do not occupy a volume of their own.
   C) The kinetic-molecular theory states that the temperature in Kelvin of the gas is dependent on the average kinetic energy of that gas.
   D) Only A and B are true
   E) A, B and C are all true statements.

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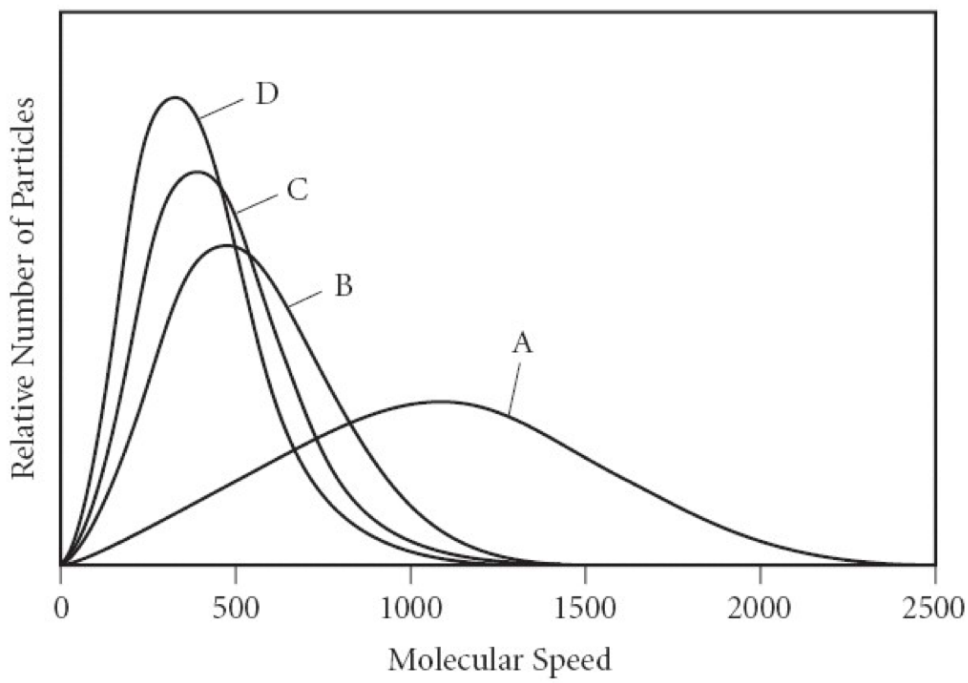
3. The following graph represents four different gases at the same temperature. Which of the following statements are true?



A) Molecules of the gas represented by curve A have a greater distribution of speeds due to the gas having a larger molar mass.
B) Molecules of the gas represented by curve C have a smaller molar mass than the molecules represent by gas B.
C) All of the curves represented are actually the same molecule and at the same temperature, but have different root mean square speeds.
D) Molecules of the gas represented by curve D have the smallest root mean square speed and would diffuse the fastest.
E) None of the above statements are true.

4. The molar heat capacity of a compound with the formula C_2H_6SO is 88.0 J/mol-K. The specific heat of this substance is__________ J/g-K.

A) 6.88×10^3
B) 1.13
C) 4.89
D) 88.0
E) -88.0

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5. Which of the following conditions will always result in an increase in the internal energy of the system?

A) system loses heat and does work on the surroundings
B) system gains heat and does work on the surroundings
C) system loses heat and has work done on it by the surroundings
D) system gains heat and has work done on it by the surroundings
E) none of the above

6. The value of $\Delta H°$ for the reaction below is -72 kJ. How many kJ of heat are released when 1.0 mol of HBr is formed in this reaction.

$$H_2\ (g) + Br_2\ (g) \rightarrow 2HBr\ (g)$$

A) -144
B) 144
C) 72
D) 0.44
E) 36

7. The value of $\Delta H°$ for the following reaction is 177.8 kJ. The value of $\Delta H_f°$ for CaO(s) is ___________ kJ/mol.

$$CaCO_3\ (s) \rightarrow CaO\ (s) + CO_2\ (g)$$

Substance	$\Delta H_f°$ (kJ/mol)
CO (g)	-110.5
CO_2 (g)	-393.7
$CaCO_3$ (s)	-1207.0

A) -1600
B) -635.5
C) 813.4
D) 177.8
E) -813.4

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8. Given the following reactions

$H_2S(g) + 2O_2(g) \rightarrow H_2SO_4(l)$	ΔH = -235.5 kJ
$H_2S(g) + 2O_2(g) \rightarrow SO_3(g) + H_2O(l)$	ΔH = -207.0 kJ
$H_2O(l) \rightarrow H_2O(g)$	ΔH = 44.0 kJ

What is the enthalpy of the following reaction?

$$H_2SO_4(l) \rightarrow SO_3(g) + H_2O(g)$$

A) - 388.5 kJ
B) -109
C) +28.5 kJ
D) +72.5 kJ
E) -15.5

9. 1.150 g of sucrose, $C_{12}H_{22}O_{11}$ goes through combustion in a bomb calorimeter. If the temperature increased from 23.42°C to 27.64°C and the heat capacity of the calorimeter is 4.90 kJ/°C what is enthalpy of combustion in kJ/mol of sucrose?

A) -20.68 kJ/mol
B) -6.152×10^3 kJ/mol
C) 23.78 kJ/mol
D) 17.98 kJ/mol
E) 6100 kJ/mol

10. What is the wavelength of light in meters that has a frequency of 1.20×10^{13} s^{-1}?

A) 25.0 m
B) 2.5 m
C) 2.50×10^{-5} m
D) 0.0400 m
E) 12.0 m

Multiple Choice - Circle the BEST Answer

11. What is the wavelength of a photon with energy of 3.7×10^{-18} J?

A) 5.4×10^{-8} m
B) 2.2×10^{3} m
C) 1.9×10^{7} m
D) 1.9×10^{-25} m
E) 3.7×10^{-21} m

12. What is the wavelength (in m) of an electron whose velocity is 1.7×10^{4} m/s and whose mass is 9.1×10^{-31} kg?

A) 12 m
B) 4.3×10^{-11} m
C) 4.3×10^{-8} m
D) 2.3×10^{-7} m
E) 2.3×10^{7} m

13. Of the following transitions in the Bohr hydrogen atom, which transition results in the emission of the highest energy photon?

A) n=3 → n=5
B) n=5 → n=1
C) n=5 → n=3
D) n=1 → n=5
E) n=1 → n=4

14. What are the quantum numbers for the outermost electron in a strontium atom (Sr) in the ground state assuming convention? (n, l, m_l, m_s)

A) 4, 3, 3, +1/2
B) 3, 1, -1, -1/2
C) 3, 0, 0, +1/2
D) 5, 0, 0, -1/2
E) 5, 1, 0, +1/2

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15. How many orbitals are present in the third energy level of an atom?

A) 2
B) 3
C) 4
D) 9
E) 10

16. How many p-orbitals are occupied in the ground state of silicon (Si)?

A) 3
B) 4
C) 5
D) 6
E) 7

17. In a ground state iron atom the ____________ subshell is partially filled.

A) 3s
B) 3p
C) 3d
D) 4s
E) 4p

18. In which orbital does an electron in an atom experience the greatest nuclear charge?

A) 3d
B) 3p
C) 3s
D) 2p
E) 2s

Multiple Choice - Circle the BEST Answer

19. Which of the following correctly represents the first ionization energy of bromine?

A) $Br\ (g) \rightarrow Br^{+}(g) + e^{-}$
B) $Br_2\ (g) + 2e^{-} \rightarrow 2\ Br^{-}\ (g)$
C) $Br\ (g) + e^{-} \rightarrow Br^{-}\ (g)$
D) $Br^{+}\ (g) + e^{-} \rightarrow Br\ (g)$
E) $Br_2\ (g) + e^{-} \rightarrow Br^{-}\ (g)$

20. Of the following species which has the largest radius?

A) Rb^{+}
B) Sr^{2+}
C) Br^{-}
D) Kr
E) Ar

21. Which form of the exam do you have?

A) A
B) B